

Topology-Synchronous Decoupled Observations and Supervised-Reset Estimation for Online C–ESR Monitoring of the Ćuk Energy-Transfer Capacitor

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Abstract—The energy-transfer capacitor in a Ćuk converter is a reliability-critical power-transfer element, but ripple measurements couple its capacitance and equivalent series resistance (ESR). This paper proposes topology-synchronous, direction-decoupled observations that exploit the current commutation between $+i_{L1}$ and $-i_{L2}$ without diagnostic injection: timestamped fits in guarded edge windows reconstruct an ESR-dominant discontinuity, and in-state charge integration yields a capacitance-dominant row. Positive finite-window information bounds establish local identifiability under valid continuous-conduction-mode conditions. On these observations, a supervised-reset Kalman estimator (TS-SRKE) augments the Joseph-form kernel with a predeclared two-time-scale supervisor over accepted and gate-rejected innovations, restoring gain after abrupt health changes without disturbing interval calibration. Evaluation covers 48 blind cases across four noise profiles and four timing skews, with abrupt-step, degradation-ramp, and TMS320F28379D device-realistic scenarios. Simulation results show that the proposed observations alone reduce mean capacitance error from 7.0366% to 0.3727% with the RLS kernel unchanged, and that TS-SRKE attains 0.3011%/0.6446% capacitance/ESR error with 88.93% joint coverage while reproducing its parent exactly in steady state. Abrupt capacitance and ESR steps are settled within 1.52 and 0.48 ms, where the unsupervised kernel does not recover; worst 95th-percentile errors of 1.6425%/2.6109% support device-realistic feasibility under the modeled DSP constraints.

Index Terms—Ćuk converter, capacitor condition monitoring, change detection, energy-transfer capacitor, capacitance estimation, equivalent series resistance, Kalman filters, parameter identification, power electronics reliability.

I. INTRODUCTION

Capacitors are reliability-critical components of power electronic converters, and their condition strongly affects converter availability [1]. However, capacitors degrade under sustained electrical and thermal stress, and the resulting capacitance loss and equivalent series resistance (ESR) growth in turn alter electrical stress, thermal loss, and filtering performance; both quantities are established degradation precursors for electrolytic and film capacitors [1], [2]. Since neither parameter is directly measurable during operation, many online monitoring methods have been developed, spanning ripple analysis, injected excitation, circuit reconfiguration, observers, recursive identification, and data-driven processing [3]–[5]. The suitability of a given scheme depends on the capacitor’s circuit role and on which electrical variables are already measured.

The energy-transfer capacitor in a Ćuk converter is distinct from a dc-link or output-filter capacitor: it lies in the main power-transfer path and alternately carries the full inductor

currents $+i_{L1}$ and $-i_{L2}$. Every switching transition therefore commutates the capacitor current between two signed values, and normal operation supplies this bidirectional excitation at the switching frequency without any diagnostic perturbation. Treating this component as a generic two-terminal capacitor sampled asynchronously discards this topology-specific information.

Existing joint C–ESR identification methods approach the problem from other circuit roles. Hasegawa *et al.* [6] estimate dc-link capacitance and ESR by separating ripple-frequency components in an inverter dc link; however, the separation is tied to the ripple bands of that topology. Yao *et al.* [7] identify ESR and capacitance of a buck output capacitor from the pulsewidth-modulation (PWM) signal and output voltage without a current sensor, and related sensorless formulations extend this route [8]; however, the underlying within-cycle relation presumes the unidirectional filter current of the buck output stage and does not carry over to a capacitor whose current reverses in every subinterval. Observer-based schemes [9]–[11] obtain joint estimates through a switched resistive network; however, they require an added discharge network or restrict monitoring to quasi-online intervals. Ribeiro *et al.* [12]–[14] extract inherent or injected spectral features for neutral-point-clamped (NPC) dc-link capacitors; however, injection adds loss, and the inherent-signal mapping is specific to the dc-link spectrum. Recent extensions include sensorless boost ESR estimation, optimized injection for buck capacitors, and large-signal C–ESR estimation [15]–[17]. Experimentally validated dc-link monitoring continues to advance through maximum-likelihood traction-inverter estimation, adaptive-forgetting recursive least squares (RLS) for cascaded-H-bridge converters, discharge-profile and grey-box transient methods, multilevel-converter ESR–C schemes, and DSP-efficient spectral precursors [18]–[24], and wavelet/Kalman and RLS/Kalman formulations provide further estimation routes [25]–[28]. For the Ćuk converter itself, fractional-order modeling, converter fault diagnosis, sensor-fault detection, and prognostics have been studied [29]–[32]; however, none of these works exploits the transfer-capacitor current commutation for joint C–ESR identification.

The noninvasive joint schemes of [6], [7], [12], [25] and the observer route of [9] are the closest related methods; the comparison remains categorical rather than numerical because the converter topologies, sensing sets, and reporting protocols differ across studies. Relative to all of them, this work is distinguished by exploiting the $+i_{L1} \leftrightarrow -i_{L2}$ commutation of

the energy-transfer capacitor itself, with the terminal voltage, the two inductor currents, and the switching state and timestamp as the only measurements. The residual gap is structural rather than algorithmic: no reported method converts the Ćuk transfer mechanism itself into implementable identification information.

Therefore, in this paper, topology-synchronous decoupled observations are proposed that perform exactly this conversion: local fits to timestamped samples in guarded pre- and post-edge windows reconstruct the ESR-dominant terminal-voltage discontinuity at the switching instant, and an ESR-compensated charge balance inside one switching state yields a capacitance-dominant row. In the common-kernel factorial of Section VII, the proposed observations alone reduce mean capacitance error from 7.0366% to 0.3727% with the RLS kernel unchanged. On these observations, a topology-synchronous supervised-reset Kalman estimator (TS-SRKE) is constructed, which restores gain after abrupt health changes while leaving steady-state behavior unchanged. The design is evaluated on 48 blind cases across four noise profiles and four timing skews, with abrupt-step, degradation-ramp, closed-loop-regulation, light-load, and TMS320F28379D device-realistic scenarios.

The main contributions of this paper are as follows.

- 1) Topology-synchronous decoupled observations that interpret the normal Ćuk switching sequence as topology-induced health excitation and convert it, through timestamp-reconstructed edge fits and safe-window charge balances, into ESR-dominant and capacitance-dominant rows.
- 2) Joint finite-window identifiability established from the raw observation pair, with positive per-direction information bounds for the implemented sequential stream, under valid continuous-conduction-mode conditions.
- 3) An observation–estimator factorial inside a predeclared blind verification protocol (48 cases, a conditional Cramér–Rao benchmark and sensitivity analysis, and a TMS320F28379D device-realistic simulation) demonstrating that the observation design, rather than added estimator complexity, carries the dominant accuracy gain.
- 4) A supervised-reset extension (TS-SRKE) that restores gain after abrupt health changes through a predeclared two-time-scale supervisor over the accepted *and* gate-rejected normalized innovations; its components are classical, and the claimed novelty is confined to their direction-specific construction on the gated topology-synchronous rows (Section V-C).

The rest of this paper is organized as follows. Section II develops the switched model of the energy-transfer capacitor and its health signatures. Section III presents the topology-synchronous decoupled observation framework. Finite-window identifiability and information bounds are established in Section IV. Section V presents the estimation kernels and the supervised-reset realization. Section VI describes the verification methodology. In Section VII, simulation results are reported. Section VIII assesses device-realistic feasibility on the

TABLE I
PRINCIPAL NOTATION

Symbol	Meaning
u, D, T_s	switch state, duty ratio, switching period
i_1, i_2	input- and output-inductor currents
i_C	reconstructed energy-transfer-capacitor current
v_C, v_T	ideal internal and measured terminal voltages
C, C_b	actual and fixed nominal capacitances
r_C	capacitor ESR at the measurement conditions
$\bar{\alpha}$	normalized inverse capacitance, C_b/C
q	charge transferred inside an accepted window
I_Σ	edge-current sum, $i_1 + i_2$
k_R	training-calibrated gain held fixed during estimation

TMS320F28379D. Section IX discusses scope and limitations. Finally, Section X concludes the paper.

II. ENERGY-TRANSFER CAPACITOR MODEL AND HEALTH SIGNATURES

A. Topology, States, and Sign Convention

Consider the classical inverting Ćuk converter in continuous conduction mode (CCM), shown in Fig. 1(a); Table I summarizes the principal notation. The state vector is

$$\mathbf{x} = [i_1 \quad i_2 \quad v_C \quad v_o]^T, \quad (1)$$

where $i_1 = i_{L1} > 0$ and $i_2 = i_{L2} > 0$ follow their usual inductor-current directions, v_C is the internal voltage of the ideal element in the energy-transfer capacitor, and $v_o > 0$ is the magnitude of the physically negative output voltage. Let $u \in \{0, 1\}$ denote the main-switch state, with $u = 1$ for the on-state.

The transfer-capacitor current is

$$i_C = (1 - u)i_1 - ui_2, \quad (2)$$

and therefore, matching the two switch topologies in Fig. 1(b) and (c),

$$u = 0 : i_C = i_1, \quad u = 1 : i_C = -i_2. \quad (3)$$

Equation (2) is the topology-specific reconstruction used throughout this work; it avoids a dedicated series current sensor in the capacitor branch when the switching state and both inductor currents are available.

B. Nonideal Capacitor Port

The first-order capacitor model consists of an ideal capacitance C in series with the ESR r_C :

$$v_T = v_C + r_C i_C, \quad \dot{v}_C = \frac{i_C}{C}, \quad (4)$$

where v_T is the externally measured terminal voltage. Equating v_T with v_C would attribute the ESR drop to the ideal capacitance and couple the two health indicators.

For numerical conditioning, a fixed nominal capacitance C_b defines

$$\bar{\alpha} = \frac{C_b}{C}, \quad C = \frac{C_b}{\bar{\alpha}}. \quad (5)$$

The health-parameter vector is $\boldsymbol{\theta} = [\bar{\alpha}, r_C]^T$.

TABLE II
CONVERTER AND IDENTIFICATION PARAMETERS

Item	Value
Input voltage	$V_{in} = 24$ V nominal; 19.2/24/28.8 V blind range
Duty ratio	$D = 0.40$ nominal; 0.30/0.40/0.55/0.65 blind range
Switching frequency	$f_s = 50$ kHz ($T_s = 20$ μ s)
Inductors	$L_1 = L_2 = 500$ μ H
Capacitors	$C_b = 100$ μ F; $C_o = 470$ μ F
Nominal ESR/load	$r_b = 50$ m Ω ; $R = 10$ Ω
Primary recursion	$\lambda = 0.9975$; $P_0 = 1000$
Edge calibration	$k_R = 0.9772$; training-calibrated nominal value; case-specific cali- brated value used in evaluation
C charge window	$T_{w,C} = 2.0$ μ s
ESR edge-fit win- dow	$T_{w,R} = 2.2$ μ s in the F28379D realization
Health reporting	1024 cycles = 20.48 ms per external report

C. Health Scope

Capacitance loss and ESR growth are treated as condition indicators, not as remaining-useful-life predictions. Because r_C depends on temperature, frequency, operating point, and aging, the method estimates effective ESR under the specified switching and acquisition conditions. The theory assumes recurring valid CCM windows, negligible parameter change within an identification window, and a first-order series C–ESR model. The constancy assumption is mild because an identification window spans tens of milliseconds whereas wear-out drift evolves over weeks to months [1]; the abrupt-change scenarios of Section VII probe its violation directly. Equivalent series inductance (ESL), ringing, discontinuous conduction mode (DCM), and front-end nonidealities are addressed through safe-window selection, calibration, gating, or explicit limitations.

Only inverse capacitance is normalized: $\bar{\alpha} = C_b/C$ is dimensionless, whereas r_C is estimated directly in ohms. Separate scalar recursions and physical bounds avoid mixing their numerical scales. The predeclared ESR interval is $0.35r_b \leq r_C \leq 2.50r_b$, or 17.5–125 m Ω for $r_b = 50$ m Ω . The converter and identification parameters used throughout the evaluation are listed in Table II.

The identification uses the terminal voltage, both inductor-current measurements, and the switching state with synchronized timestamps; applications lacking either inductor-current measurement are addressed in Section IX. The following section converts these measurements into decoupled health information.

III. TOPOLOGY-SYNCHRONOUS DECOUPLED OBSERVATION FRAMEWORK

In this section, the commutation of Section II is converted into two decoupled observation rows. The natural excitation is established first; the timestamp-reconstructed ESR row and the safe-window charge row follow, together with their measurement covariances and the validity gates shared by all compared estimators. Fig. 2 illustrates the excitation and both

observation windows; the complete estimation loop built on these rows is presented in Section V.

A. Natural Bidirectional Excitation

At a valid $u : 0 \rightarrow 1$ transition, the capacitor current changes from $i_C^- = i_1^-$ to $i_C^+ = -i_2^+$. Hence

$$\Delta i_C = i_C^+ - i_C^- = -(i_1^- + i_2^+). \quad (6)$$

Because the internal ideal-capacitor voltage cannot jump, $v_C^+ = v_C^-$. The terminal voltages immediately before and after the edge satisfy

$$v_T^- = v_C + r_C i_1^-, \quad (7)$$

$$v_T^+ = v_C - r_C i_2^+, \quad (8)$$

which gives the ideal ESR signature

$$v_T^- - v_T^+ = r_C (i_1^- + i_2^+). \quad (9)$$

Thus, the topology transition provides a large, signed current change while canceling the continuous internal voltage.

B. Timestamp-Reconstructed ESR Observation

Direct edge samples are contaminated by acquisition skew, finite bandwidth, and switching transients. The proposed measurement therefore fits local linear models to timestamped points in safe pre- and post-edge windows and extrapolates both models to a common switching timestamp, estimating the discontinuity at one physical instant rather than subtracting samples taken at different offsets. With reconstructed intercepts $\hat{v}_T^-$ and $\hat{v}_T^+$, the accepted ESR row is

$$z_R = \hat{v}_T^- - \hat{v}_T^+ = k_R I_{\Sigma} r_C + \nu_R, \quad I_{\Sigma} = i_1^- + i_2^+, \quad (10)$$

where k_R is a training-locked calibration coefficient and ν_R collects voltage, timestamp, fitting, and residual front-end errors.

The absence of diagnostic injection does not imply zero sensing burden. While the measurement set of Section II adds no sensors, the device-realistic realization in Section VIII uses separate absolute- and edge-voltage ranges in the differential v_T analog front end. No capacitor-branch current sensor is required.

Simulation calibration and blind health identification use disjoint cases. Four calibration profiles—nominal, high duty/load, high noise, and nominal parasitic Model B—form the accepted-edge ratio

$$\kappa_m = \frac{z_{R,m}}{I_{\Sigma,m} r_{C,\text{ref},m}}, \quad \hat{k}_R = \text{median}_m \kappa_m, \quad (11)$$

where $r_{C,\text{ref},m}$ is the planted simulation reference ESR. The resulting training-locked coefficient is $k_R = 0.9772$, reported to four decimal places; the full-precision value used in all regenerated results is retained in Supplementary Note S2 and the source-traceability record.

In the evaluation, a declared ± 0.006 case-wise spread about $\hat{k}_R$ represents unit-to-unit variation in the calibrated measurement-chain gain. For each simulated case, the realized coefficient is assumed known after calibration and is supplied identically to all compared estimators through $h_R = k_R I_{\Sigma}$.

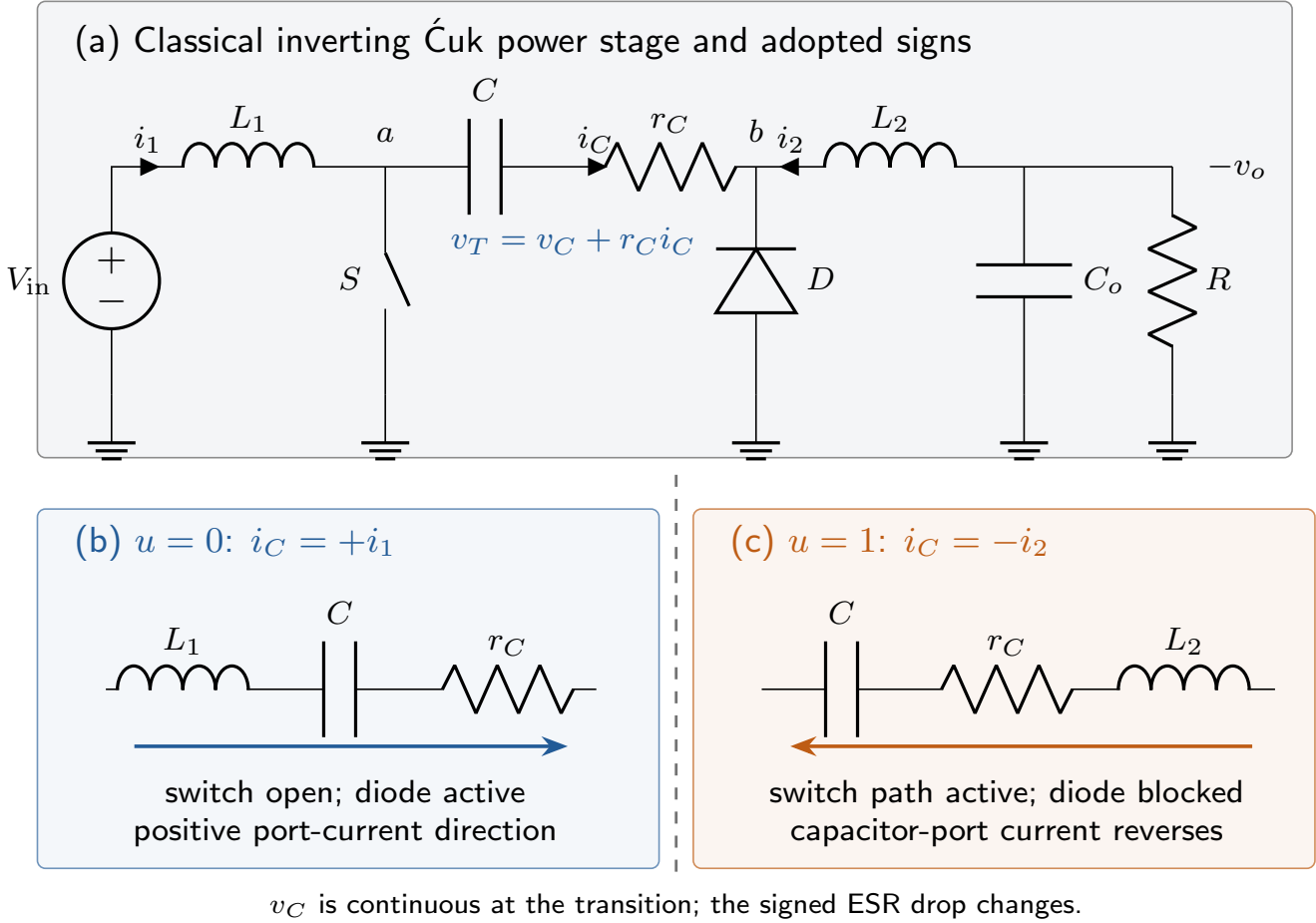


Fig. 1. Classical inverting Ćuk converter and adopted current/voltage signs: (a) full power stage; (b) $u = 0$ topology with $i_C = +i_1$; (c) $u = 1$ topology with $i_C = -i_2$. The ideal capacitor voltage v_C is continuous at a transition, while the terminal voltage $v_T = v_C + r_C i_C$ contains the ESR step.

The spread therefore tests calibrated unit-to-unit variability rather than residual unknown calibration error; k_R is never updated from blind residuals. A hardware realization would require commissioning with an independently characterized reference capacitor under the final measurement chain.

Residual calibration error is characterized analytically. If the estimators are supplied $\hat{k}_R = (1 + \delta)k_R$ while the data carry k_R , the edge row identifies $r_C/(1 + \delta)$ in the noise-free limit: a fixed relative calibration error δ maps into a multiplicative ESR bias of $-\delta/(1 + \delta)$. Through the compensation term in (15), the same error adds a capacitance-row contribution of about $\delta r_C \Delta i_C$, bounded relative to the charge term by $\delta |r_C \Delta i_C| C_b / (|q| \bar{\alpha})$. Because a fixed scale cancels in the ratios $\hat{r}_C / \hat{r}_{C,0}$ and $\hat{C} / \hat{C}_0$, health trending tolerates a constant residual calibration error, whereas absolute ESR retains the commissioning requirement above.

Thus, k_R is known and fixed within each identification interval; otherwise the edge row identifies only the product $k_R r_C$. The smaller evaluated-envelope value $k_{R,\min}^{\text{eval}} = 0.957654$ is used only for a conservative information bound and does not resolve this scale ambiguity.

C. Safe-Window Charge Observation

For an interval $[t_a, t_b]$ wholly inside one switching state, with the charge-safe-window length denoted by $T_{w,C} = t_b - t_a$, define

$$q = \int_{t_a}^{t_b} i_C(t) dt. \quad (12)$$

Integrating (4) over this interval gives

$$\Delta v_T = \frac{q}{C} + r_C \Delta i_C. \quad (13)$$

Using the most recent accepted ESR estimate gives the capacitance pseudo-measurement

$$z_C = \Delta v_T - \hat{r}_C \Delta i_C = \frac{q}{C_b} \bar{\alpha} + v_C. \quad (14)$$

Table III summarizes the two rows. ESR-estimation uncertainty and residual correlation enter the conditional measurement covariance.

More explicitly, first-order substitution of measured end-point current, voltage, charge, and the preceding ESR estimate gives

$$z_C = \frac{q}{C_b} \bar{\alpha} + (r_C - \hat{r}_C) \Delta i_C + v_{\Delta v} - \hat{r}_C v_{\Delta i} + v_q, \quad (15)$$

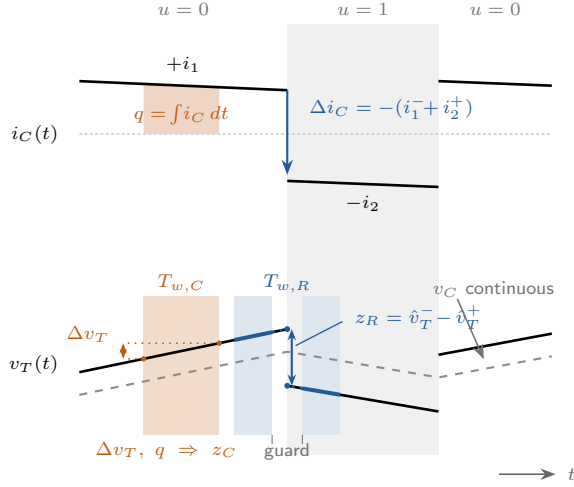


Fig. 2. Topology-synchronous excitation and decoupled observations over 1.5 switching periods (schematic; guard exaggerated for readability). The capacitor current commutates between $+i_1$ and $-i_2$; the internal voltage v_C (dashed) is continuous while the terminal voltage v_T carries the ESR discontinuity, reconstructed by local fits in the guarded windows $T_{w,R}$ extrapolated to the common edge time. The charge integral q and endpoint difference Δv_T over $T_{w,C}$ form the capacitance row.

TABLE III
DECOUPLED OBSERVATION ROWS

Row	Measurement	Regression	Dominant parameter
Edge	$\hat{v}_T^- - \hat{v}_T^+$	$k_R I_\Sigma r_C + \nu_R$	r_C
Charge	$\Delta v_T - \hat{r}_C \Delta i_C$	$(q/C_b)\bar{\alpha} + \nu_C$	$\bar{\alpha}$

where ν_q is the voltage-equivalent charge-integration error. In the evaluated implementation its unscaled charge variance is denoted R_q , so that $\text{var}(\nu_q) = \bar{\alpha}^2 R_q$. With the preceding ESR error treated as a zero-mean conditional uncertainty of variance P_{rr} , the conditional measurement covariance can be approximated as

$$R_C \simeq R_{\Delta v} + (\Delta i_C)^2 P_{rr} + \hat{r}_C^2 R_{\Delta i} + \bar{\alpha}^2 R_q. \quad (16)$$

Here $R_{\Delta v} = 2R_v$, $R_{\Delta i} = 2R_i$, and R_q follows from current-noise variance, the analog-to-digital converter (ADC) interval, and the integration length. Equation (16) propagates uncertainty; a biased $(r_C - \hat{r}_C)\Delta i_C$ term remains bounded model mismatch rather than white noise.

D. Validity and Update Policy

An observation is accepted only in CCM with a valid timestamp, topology state, and complete safe window. Denoting these prechecks by $\mathcal{S}_C$ and $\mathcal{S}_R$, the common evaluation stream implements

$$\begin{aligned} \mathcal{V}_C &= \text{CCM} \wedge \mathcal{S}_C \wedge (n_C > 0), \\ \mathcal{V}_R &= \text{CCM} \wedge \mathcal{S}_R \wedge (I_\Sigma \geq I_{\Sigma, \text{gate}}), \end{aligned} \quad (17)$$

with the predeclared $I_{\Sigma, \text{gate}} = 0.12$ A. Charge acceptance uses the CCM/safe-window flag and a nonzero accepted-charge count; the implementation has no separate numeric

q_{gate} . A normalized-innovation-squared (NIS) threshold of 9 may reject additional TS-SLTVKE outliers. A failed gate holds the corresponding estimate. The evaluated-envelope minima in Section IV are proof constants, not online thresholds.

Because the capacitance regressor $h_C = q/C_b$ vanishes with the accepted charge, the positive information guarantee is restricted to the evaluated valid-CCM envelope rather than asserted for arbitrarily small q .

IV. FINITE-WINDOW IDENTIFIABILITY AND INFORMATION BOUNDS

A. Joint Identifiability from the Raw Observation Pair

Identifiability is established on the raw observations, before any sequential ESR compensation. Stacking the uncompensated charge relation (13), written as $\Delta v_T = (q/C_b)\bar{\alpha} + r_C \Delta i_C + \nu_{\Delta v}$, with the edge row (10) gives

$$\begin{bmatrix} \Delta v_T \\ z_R \end{bmatrix} = \mathbf{H}_k \boldsymbol{\theta} + \boldsymbol{\nu}_k, \quad \mathbf{H}_k = \begin{bmatrix} q/C_b & \Delta i_C \\ 0 & k_R I_\Sigma \end{bmatrix}. \quad (18)$$

Since $\det \mathbf{H}_k = (q/C_b)k_R I_\Sigma$, the matrix has full rank whenever $q \neq 0$, $I_\Sigma \neq 0$, and $k_R \neq 0$. These conditions hold over the evaluated valid-CCM envelope: the edge-current condition is enforced explicitly by (17), nonzero accepted charge follows from the sign-invariant CCM charge windows of that envelope, and $k_R \neq 0$ is a fixed calibration assumption rather than an online gate. For finite positive-definite raw-observation covariances $\mathbf{R}_k$, the joint Gramian $\sum_k \mathbf{H}_k^T \mathbf{R}_k^{-1} \mathbf{H}_k$ is therefore positive definite for any window containing at least one complete accepted pair. Conditional on the accepted reconstructed regressors and the fixed calibration coefficient, $(\bar{\alpha}, r_C)$ is jointly locally identifiable from the raw observation model; since $C = C_b/\bar{\alpha}$ is locally one-to-one for $\bar{\alpha} > 0$, the same holds for (C, r_C) . No estimate of r_C enters this claim. The stacked pair is an identifiability construction only: the online implementation need not process the two rows simultaneously, and the sequential bounds below count m_C charge and m_R edge rows separately.

B. Information Structure of the Sequential Implementation

The implemented recursion processes the triangular structure of (18) sequentially: the edge row involves r_C alone, and the charge row is compensated with the most recent ESR estimate in (14), whose residual $(r_C - \hat{r}_C)\Delta i_C$ is propagated in the conditional covariance (16) rather than ignored. For this decoupled stream, recall the regressors $h_C = q/C_b$ and $h_R = k_R I_\Sigma$, with conditional variances R_C and R_R ; the per-direction finite-window information Gramian is

$$\mathbf{G}_{\theta, N} = \sum_{k \in \mathcal{K}_C} \frac{1}{R_{C,k}} \begin{bmatrix} h_{C,k}^2 & 0 \\ 0 & 0 \end{bmatrix} + \sum_{k \in \mathcal{K}_R} \frac{1}{R_{R,k}} \begin{bmatrix} 0 & 0 \\ 0 & h_{R,k}^2 \end{bmatrix}. \quad (19)$$

Suppose that on every accepted charge interval the current satisfies $|i_C| \geq I_{C, \text{min}}^{\text{eval}}$ without changing sign and that the interval length is at least $T_{w, C, \text{min}}$. Then $|q| \geq q_{\text{min}}^{\text{eval}} = I_{C, \text{min}}^{\text{eval}} T_{w, C, \text{min}}$. If $R_C \leq R_{C, \text{max}}$ and $R_R \leq R_{R, \text{max}}$, if the calibrated k_R stays fixed with a known conservative lower bound $k_{R, \text{min}}^{\text{eval}} > 0$ over the evaluated envelope, and

if $I_\Sigma \geq I_{\Sigma,\min}^{\text{eval}} > 0$, then the per-direction information of a window containing m_C charge rows and m_R edge rows is bounded below by

$$\underline{\mu}_C = m_C \frac{(I_{C,\min}^{\text{eval}} T_{w,C,\min}/C_b)^2}{R_{C,\max}}, \quad (20)$$

$$\underline{\mu}_R = m_R \frac{(k_{R,\min}^{\text{eval}})^2 (I_{\Sigma,\min}^{\text{eval}})^2}{R_{R,\max}}. \quad (21)$$

These bounds yield the following result for the implemented stream.

Proposition 1 (Direction-wise finite-window information positivity): Assume valid CCM; k_R known and fixed during each identification window; sign-invariant accepted charge windows; positive accepted edge-current sum; finite conditional covariances; and at least one accepted charge and edge row in a finite window. Then

$$\mathbf{G}_{\theta,N} \succeq \text{diag}(\underline{\mu}_C, \underline{\mu}_R) \succ \mathbf{0}, \quad (22)$$

so the implemented sequential stream retains strictly positive finite-window information in both parameter directions.

Proof: Each charge row contributes a positive semidefinite matrix only in the $\bar{\alpha}$ direction, and each edge row contributes one only in the r_C direction. Applying the physical lower bounds in (20) and (21), summing m_C and m_R accepted contributions, and noting that both lower bounds are strictly positive establishes (22). $\square$

Remark 1 (Conditional nature of the excitation): PWM alone does not guarantee persistent excitation (PE) at every load or conduction mode. The Ćuk topology supplies the physical directions, while accepted rows certify when the bound applies. The superscript “eval” marks minima over the evaluated CCM envelope; these constants are neither online gates nor post-hoc tuning values.

C. Numerical Lower-Bound Check

The constants of Proposition 1 are instantiated on the evaluated envelope as follows. Across 45 independent Model-B CCM cases, the weakest physical quantities are

$$\begin{aligned} I_{C,\min}^{\text{eval}} &= 0.330108 \text{ A}, & I_{\Sigma,\min}^{\text{eval}} &= 0.438504 \text{ A}, \\ q_{\min}^{\text{eval}} &= 0.660217 \text{ } \mu\text{C} \end{aligned} \quad (23)$$

with $T_{w,C,\min} = 2.0 \text{ } \mu\text{s}$ and $k_{R,\min}^{\text{eval}} = 0.957654$. The resulting weakest information bounds are

$$\underline{\mu}_C = 0.217943, \quad \underline{\mu}_R = 880.008. \quad (24)$$

The minimum ratios of empirical information to the corresponding conservative bound are 1.02872 and 1.02334. Hence every retained case exceeds its computed bound, and the weakest case rather than a nominal case determines the reported constants; Fig. 3 plots the empirical information against the corresponding bounds for all retained cases. These existence bounds are complemented by the conditional CRLB benchmark of Section VII, which quantifies how closely the realized estimator variance approaches the accepted-row information.

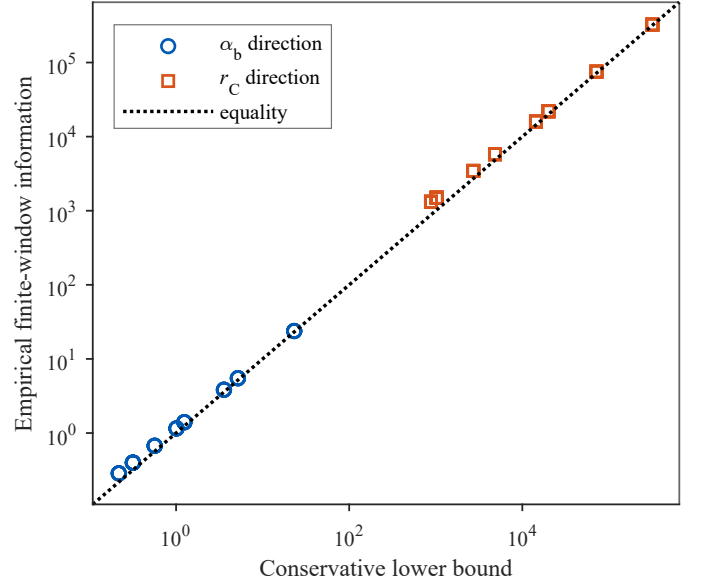


Fig. 3. Theoretical lower bounds versus empirical finite-window information over the Model-B CCM cases. Circles denote normalized inverse capacitance and squares denote physical ESR; their parameterization-dependent magnitudes and units are not directly comparable. All accepted cases lie above the equality line.

D. Covariance Contraction for the Uncertainty Extension

For the interval-reporting extension of Section V, let P_n be a posterior variance, Q_N the process variance accumulated over a finite window, and $\mu \geq \underline{\mu} > 0$ the accepted information.

Proposition 2 (Finite-window covariance boundedness): For recurring valid updates, finite $0 \leq Q_N \leq \bar{Q}_N$, and $\mu \geq \underline{\mu} > 0$, the posterior variance satisfies

$$P_{n+1} \leq \frac{P_n + Q_N}{1 + \mu(P_n + Q_N)}. \quad (25)$$

For constant Q_N , the positive fixed point of (25) is

$$P^* = \frac{-Q_N + \sqrt{Q_N^2 + 4Q_N/\mu}}{2}. \quad (26)$$

For $Q_N = 0$, $P_n^{-1} \geq P_0^{-1} + n\mu$.

Proof: See Appendix B, which also reports validated fixed-point values. $\square$

Remark 2 (Scope of the contraction result): Proposition 2 is a conditional linear-model result consistent with standard filtering theory [33], [34], not a global nonlinear-converter convergence theorem. Projection is inactive in all 200 legal-PE seeds and is treated as a safety constraint rather than the source of contraction.

V. ONLINE ESTIMATION: KERNELS AND THE SUPERVISED-RESET REALIZATION

All realizations in this section consume the same accepted edge and charge rows from Section III, separating observation effects from recursive-kernel effects. The two kernel baselines are presented first, their tracking–confidence tradeoff is isolated, and the supervised-reset realization follows.

A. Kernel Baselines

The lightweight baseline TS-D-RLS—a topology-synchronous direction-decoupled RLS—uses one scalar RLS recursion per health direction. For direction $j \in \{C, R\}$, an accepted row $z_{j,k} = h_{j,k}\theta_{j,k} + \nu_{j,k}$ updates

$$K_{j,k} = \frac{P_{j,k-1}h_{j,k}}{\lambda + h_{j,k}^2 P_{j,k-1}}, \quad (27)$$

$$\hat{\theta}_{j,k} = \hat{\theta}_{j,k-1} + K_{j,k}(z_{j,k} - h_{j,k}\hat{\theta}_{j,k-1}), \quad (28)$$

$$P_{j,k} = \lambda^{-1}(P_{j,k-1} - K_{j,k}h_{j,k}P_{j,k-1}), \quad (29)$$

with the predeclared forgetting factor $\lambda = 0.9975$; a larger λ smooths the estimate at the cost of a longer settling time. Only the direction associated with an accepted observation is updated, validity gating rejects rows that fail the predeclared acceptance rules, and projection enforces the physical parameter intervals. The reported health values are $\hat{C} = C_b/\hat{\alpha}$ and $\hat{E}SR = \hat{r}_C$.

The Kalman kernel TS-SLTVKE—a topology-synchronous scalar linear-time-varying (LTV) Kalman estimator—maintains the structured state $\mathbf{x}_k = [v_{C,k} \ \bar{\alpha}_k \ r_{C,k}]^T$ with the charge-driven transition

$$\mathbf{F}_k = \begin{bmatrix} 1 & q_k/C_b & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad (30)$$

The structured measurement directions $\mathbf{H}_V = [1, 0, 0]$, $\mathbf{H}_C = [0, q/C_b, 0]$, and $\mathbf{H}_R = [0, 0, k_R I_\Sigma]$ enter scalar Joseph-form covariance updates with predeclared process and measurement covariances, and NIS gating uses the predeclared gate $\gamma = 9$. The posterior covariance provides calibrated interval reporting. Both kernels are standard recursions [33], [34]; neither is claimed as a novelty, and the TS- names denote only their topology-synchronous, direction-specific use.

B. The Tracking–Confidence Tradeoff of the Unsupervised Kernels

The two kernels fail in complementary ways. The RLS baseline tracks and recovers quickly, but its auxiliary residual-weighted interval is uncalibrated (13.00% joint coverage in Section VII). The Kalman kernel reports near-nominal simultaneous coverage (88.93%), but after an abrupt health change every innovation exceeds the NIS gate, the update is rejected, the covariance stays small, and the estimate freezes at the stale value—the retained negative result of the unsupervised realization. The mechanism also identifies the missing information: the gate-rejected innovations, which the point update discards, carry the sustained same-sign evidence of a parameter change.

C. Primary Realization: Supervised-Reset Estimator TS-SRKE

TS-SRKE augments the Kalman kernel with a per-direction supervisor that closes the loop shown in Fig. 4. The supervisor watches the clipped normalized innovation of *every* valid row, accepted or rejected,

$$\varepsilon_{j,k} = \text{clip}\left(\frac{z_{j,k} - h_{j,k}\hat{\theta}_{j,k-1}}{\sqrt{S_{j,k}}}, \pm c\right), \quad S_{j,k} = h_{j,k}^2 P_j + R_{j,k}. \quad (31)$$

Two exponentially weighted means (EWMA) with separated time scales,

$$\begin{aligned} m_j^f &\leftarrow m_j^f + a_f (\varepsilon_{j,k} - m_j^f), \\ m_j^s &\leftarrow m_j^s + a_s (\varepsilon_{j,k} - m_j^s), \end{aligned} \quad (32)$$

act as the current innovation level and its bias reference. When $|m_j^f - m_j^s| > \tau$, the supervisor resets that direction's covariance to its initialization value, re-anchors $m_j^f \leftarrow m_j^s$, and holds off for N_h valid rows; arming requires an N_w -row learning window per direction. All constants are predeclared and shared by both directions: $a_f = 1/16$ and $a_s = 1/128$ (an eightfold time-scale separation), $\tau = 2.5$, $c = 6$, $N_w = N_h = 32$.

The two-time-scale form is deliberate. Residual timing skew leaves a deterministic, near-periodic disturbance in the edge row whose σ -normalized amplitude grows as noise shrinks; a plain CUSUM misfires on it, whereas the fast filter attenuates the zero-mean disturbance below τ while a sustained parameter-change shift crosses it within a few valid rows. After a reset the Kalman gain is restored, the estimate re-converges as from initialization, and the reported interval widens and then re-tightens, consistent with the increased post-change uncertainty. The supervisor adds four multiplications and about seven additions/comparisons per valid row to the kernel. Its components—EWMA residual filtering, change supervision, and covariance resetting—are classical [35]–[38]; the contribution is their direction-specific construction on the gated topology-synchronous rows.

VI. VERIFICATION METHODOLOGY

This section declares the models, comparisons, fairness rules, and metrics that all results in Sections VII and VIII consume; every threshold and hyperparameter is fixed before blind evaluation and never retuned afterward. All models and estimators are implemented in MATLAB (R2023b).

A. Model and Data Hierarchy

The evaluation separates equation checks, independent plant traces, estimator comparisons, and device-realistic acquisition:

- 1) Model A is a switched-equation Čuk model for sign, timing, and rapid dynamic checks.
- 2) Model B is an independently constructed Simscape model; its 45 traces calibrate current scale and edge slope and supply the blind observation data.
- 3) The estimator comparison uses 48 stratified blind operating/health cases, four noise profiles, and four residual timing-skew levels under common O1 observations.
- 4) The F28379D model adds register-realizable sampling, aperture/occupancy, reference/clock effects, analog-front-end (AFE) residuals, and Monte Carlo variation; it remains a simulation.

Supplementary closed-loop-regulation and light-load-availability studies reuse the locked pipeline on Model-A cycles, and a conditional CRLB study benchmarks estimator error against the noise-only bound; all three are reported in Section VII.

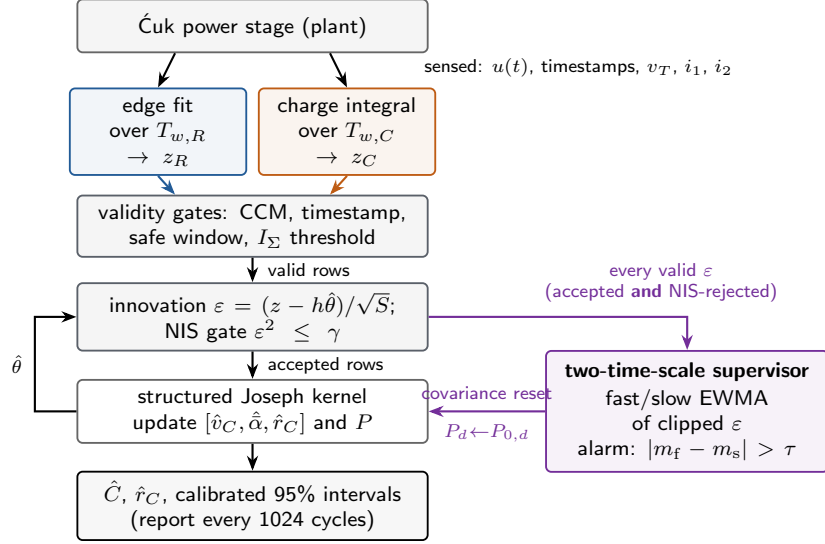


Fig. 4. The complete supervised estimation loop: per-cycle features from the sensed converter waveforms form the two observation rows; the validity gates of (17) admit valid rows; the innovation ε is built from the fed-back estimate $\hat{\theta}$ and screened by the NIS gate, and accepted rows update the structured Joseph kernel. The two-time-scale supervisor watches every valid ε —accepted or NIS-rejected—restoring gain by a per-direction covariance reset after a sustained shift; rows failing the validity gates produce no innovation and hold the estimate.

B. Factorial and Blind Comparisons

The observation–estimator factorial crosses a generic mixed-observation baseline (O0) and the proposed direction-decoupled set (O1) with RLS (E1), the dual extended Kalman filter (Dual EKF, E2), the structured LTV/Joseph estimator (E3), and the supervised-reset estimator (E4, TS-SRKE). O0 is defined explicitly: each update uses the single non-decoupled row $\Delta v_T = (q/C_b)\bar{\alpha} + \Delta i_C r_C + \nu$ built from generic within-cycle samples, retaining the raw acquisition-lag contribution and using no edge-timestamp reconstruction, so capacitance and ESR share one coupled regression. O0 is not claimed as a reproduction of any single published method; it represents the generic coupled within-cycle observation that discards topology-synchronous edge reconstruction. O1 replaces this row with the decoupled pair of Section III. Both sets use identical sensors, identical seeds, and one predeclared hyperparameter row per estimator family, so the factorial isolates the observation structure itself. All $2 \times 4 \times 768 = 6144$ paired rows are retained, and the E1–E3 rows reproduce the pre-supervision campaign exactly because the observation streams are seeded per condition and shared. The final comparison evaluates TS-SRKE against TS-D-RLS, TS-SLTVKE, and Dual EKF under O1 with predeclared primary-selection criteria: steady-state behavior within 2% of the Kalman parent, best-or-equal abrupt settling relative to the two retained realizations (TS-D-RLS and TS-SLTVKE), coverage within two percentage points of the parent, and no worse load-transient false-health peaks.

The broader baseline study includes closed-form, mixed-RLS, augmented/Dual-EKF, and Ćuk-adapted wavelet–Kalman methods. An NPC spectral method was excluded because the literature provides no unique mapping to the Ćuk edge/charge convention.

C. Fairness Rules and Metrics

All methods use identical blind cases, observations, seeds, initialization distributions, noise/timing realizations, and predeclared hyperparameters; Table IV lists the blind verification and estimator settings. Failed or saturated rows are retained, no method is retuned after blind evaluation, and paired effects use 10,000 bootstrap resamples. The resampling treats the case-condition rows as exchangeable with estimator pairing preserved; rows sharing a base plant case are not independent, so the reported intervals quantify uncertainty over the evaluated grid rather than over an independent population of converters.

Abrupt C/ESR discontinuities are estimator stress tests. The principal dynamic cases are linear and smooth C-down, ESR-up, and joint ramps lasting 0.1, 1, 10, and 100 s. Long ramps use trace-derived O1 sufficient statistics, while 0.1 s cases are cross-checked at every switching pair with Model A. Load, input-voltage, and duty transitions hold health constant to measure false-health response. None of these trajectories is hardware aging data.

Static endpoint errors are

$$e_C = 100 \left| \hat{C}_N / C - 1 \right|, \quad e_R = 100 \left| \hat{r}_{C,N} / r_C - 1 \right|, \quad (33)$$

with the mean absolute percentage error (MAPE) and 95th percentile (p95) computed over retained rows. Interval calibration is reported as joint coverage—the fraction of retained rows whose marginal 95% capacitance and ESR intervals cover their true values simultaneously; under approximate independence the nominal benchmark is $0.95^2 \approx 90.25\%$. “Convergence cycles” is the first cycle that begins a run of 32 consecutive cycles with both relative errors within 5%; unconverged cases are assigned the 1024-cycle horizon. Rejected cycles hold the estimate and remain in the count. This internal settling metric differs from the 1024-cycle (20.48-ms) external reporting cadence.

TABLE IV
BLIND VERIFICATION AND ESTIMATOR SETTINGS

Item	Predeclared setting	Role/source boundary
Health/load matrix	$C/C_b \in \{0.8, 0.9, 1.0\}$; $r_C/r_b \in \{1, 1.5, 2\}$; load factors 0.58/1/1.45	Stratified blind plant matrix; retained CCM rows only
Initialization	realized $\hat{C}_0/C \in [0.708, 1.287]$; $\hat{r}_{C,0}/r_C \in [0.512, 1.483]$	Read from the blind-case table
Parameter bounds	$C/C_b \in [0.65, 1.35]$; $r_C/r_b \in [0.35, 2.50]$	Predeclared projection limits; failed/saturated rows retained
Noise profiles	$(\sigma_v, \sigma_i) = (1 \text{ mV}, 0.5 \text{ mA}), (5 \text{ mV}, 2 \text{ mA}), (10 \text{ mV}, 5 \text{ mA}), \text{ and } (2.2 \text{ mV}, 1.2 \text{ mA})$	Shared realization; last pair is F28379D-realistic
Residual skew	0, 20, 50, and 100 ns	Shared by all compared estimators
Online validity/gating	valid CCM/state/time and safe feature; $I_{\Sigma, \text{gate}} = 0.12 \text{ A}$; no numeric q_{gate} ; TS-SLTVKE NIS gate 9	Fixed before evaluation, not inferred from PE minima
Evaluated PE constants	$I_{C, \min}^{\text{eval}} = 0.330108 \text{ A}$; $I_{\Sigma, \min}^{\text{eval}} = 0.438504 \text{ A}$; $q_{\min}^{\text{eval}} = 0.660217 \mu\text{C}$	Minima over 45 evaluated Model-B CCM cases; proof only, never online tuning thresholds
Blind comparison	48 cases $\times$ 4 noise levels $\times$ 4 skews; O0/O1 $\times$ RLS/Dual EKF/LTV-Joseph/TS-SRKE	O1 is then shared by all four estimators
Statistics/dynamics	10,000 paired bootstrap resamples; 0.1/1/10/100-s C, ESR, and joint ramps	Load, input-voltage, and duty steps hold physical health fixed
TS-SLTVKE settings	$Q_\alpha = 2 \times 10^{-9}$, $Q_r = 5 \times 10^{-9}$, $P_{0, \alpha} = 0.0144$, $P_{0, r}$ scale 0.45, NIS 9	Held fixed before blind comparison; shared by the TS-SRKE kernel
TS-SRKE supervisor	$a_f = 1/16$, $a_s = 1/128$, $\tau = 2.5$, clip 6, warm-up/hold-off 32 rows	Predeclared, shared by both directions; never retuned per case
Device acquisition	1.093 MS/s/module; 0.5- μs guard; $T_{w, R} = 2.2 \mu\text{s}$; three points/side	Constrained F28379D simulation only

For a trajectory $y(t)$ with initial truth y_0 , the implemented tracking metric is

$$\text{nRMSE}_y = \sqrt{\frac{\int (\hat{y} - y)^2 dt}{\int (y - y_0)^2 dt}}. \quad (34)$$

Combined nRMSE is the unweighted mean of finite C and ESR values. Normalized lag is the largest finite C/ESR delay estimated from zero-mean cross-correlation and from 50%-trajectory crossings, divided by the trajectory duration. Missed threshold crossings are retained but are not used to select the primary estimator. Supplementary Note S3 records the complete threshold and bootstrap definitions.

VII. RESULTS

A. Attribution to the Observation Framework

The observation–estimator factorial provides the clearest attribution. Replacing O0 with O1 while retaining the E1 RLS kernel reduces mean capacitance MAPE from 7.0366% to 0.3727%, mean ESR MAPE from 1.1970% to 0.2233%, and mean internal convergence from 665.41 to 13.21 switching cycles. The paired ESR reduction is 0.97369 percentage points (95% bootstrap interval 0.92352–1.0265); all 4608 rows are retained. Supplementary Note S3 gives the other paired intervals.

O1 also reduces capacitance MAPE by more than 6.7 percentage points for Dual EKF and E3 LTV/Joseph; Fig. 5 shows the paired effects with their bootstrap intervals. The improvement is not universal: the Dual-EKF ESR-p95 effect is negative and its interval does not favor O1. Thus, the factorial supports the observation framework without implying monotonic improvement in every estimator metric.

TABLE V
STATIC O1 ESTIMATOR COMPARISON WITH INTERNAL CONVERGENCE

Method	C MAPE (%)	ESR MAPE (%)	C/ESR p95 (%)	Conv. (int. cycles)
TS-D-RLS	0.3727	0.2233	1.3182/0.7002	13.21
TS-SLTVKE	0.3011	0.6446	1.1069/2.4992	33.09
Dual EKF	0.3043	1.1938	1.1492/3.8602	40.58
TS-SRKE	0.3011	0.6446	1.1069/2.4992	33.09

The TS-SRKE row equals its Kalman parent digit for digit: the supervisor stays silent on all 768 static blind rows.

B. Static Estimator Comparison Under O1

Table V compares the four candidates on the $48 \times 4 \times 4$ O1 matrix, and Fig. 6 visualizes the mean errors; none diverges. The supervised TS-SRKE reproduces its Kalman parent *exactly* in every static metric—the supervisor stays silent across all 768 blind rows, including the low-noise/high-skew cells that carry the largest deterministic edge disturbance. TS-D-RLS retains the lowest ESR error and fastest native convergence, a baseline strength reported throughout. Under the predeclared selection criteria of Section VI, TS-SRKE is adopted as the primary realization once the dynamic results below are included; TS-D-RLS is retained as the lightweight tracking alternative.

C. Noise and Timing Robustness

All combinations of four noise profiles and 0, 20, 50, and 100 ns residual timing skew are retained. At 50 ns under F28379D-realistic noise, C/ESR p95 errors are 0.4964%/0.3385% for TS-D-RLS, 0.4132%/0.6888% for TS-SLTVKE, and 0.4211%/1.9141% for Dual EKF; the TS-SRKE rows coincide with TS-SLTVKE at every noise/skew cell. Figure 7 visualizes that device-realistic profile; C and

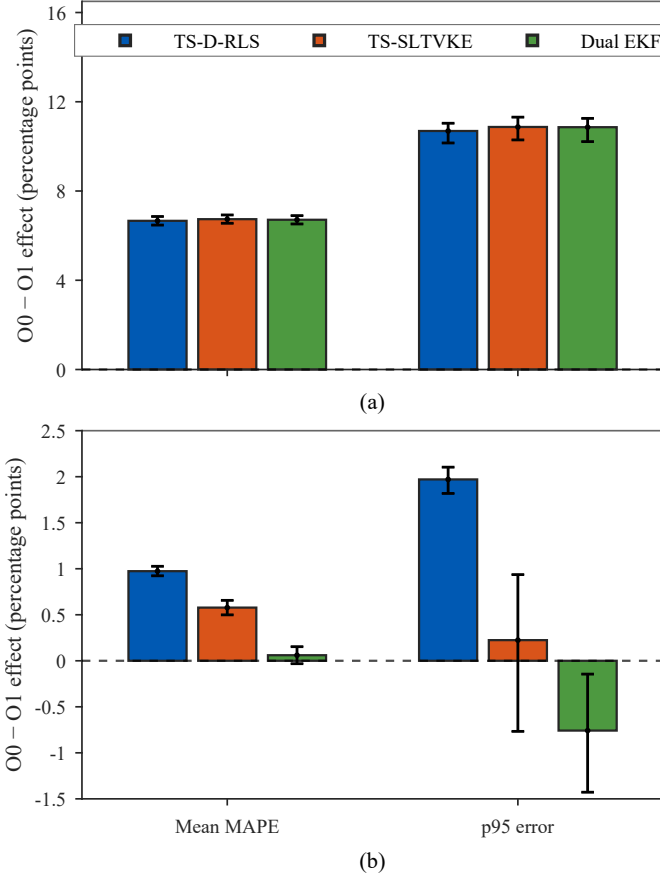


Fig. 5. Paired O0-O1 effects on MAPE and p95 error with bootstrap intervals: (a) capacitance; (b) ESR. Positive values favor the topology-decoupled O1 observations; the adverse Dual-EKF ESR-p95 effect is retained.

ESR percentiles are shown separately because their arithmetic mean has no independent interpretation.

D. Conditional CRLB Benchmark

A conditional CRLB is computed from the accepted-row information of the structured estimator lineage (Section V) on Model-A traces through the register-nominal measurement chain, over fifteen load, duty, ADC-rate, AFE-bandwidth, and timing sensitivity cases with twenty seeds each. When the chain matches its commissioned configuration, the empirical RMSE stays within factors 1.38–2.34 (capacitance) and 1.35–3.27 (ESR) of the bound; the medians over all fifteen cases are 2.17 and 2.40 (Fig. 8). Deviations from the commissioned chain instead appear as bias that the noise-only bound does not model: a reduced 1.5-MHz AFE bandwidth raises the capacitance ratio to 25.0, an increased 3-MHz bandwidth raises the ESR ratio to 10.2, and 100/200-ns residual timing skew raises the capacitance ratio to 13.0/39.6. The bound therefore serves as a conditional information benchmark: noise-driven error stays within small multiples of it at the commissioned configuration, while deviations from that configuration flag calibration or timing bias.

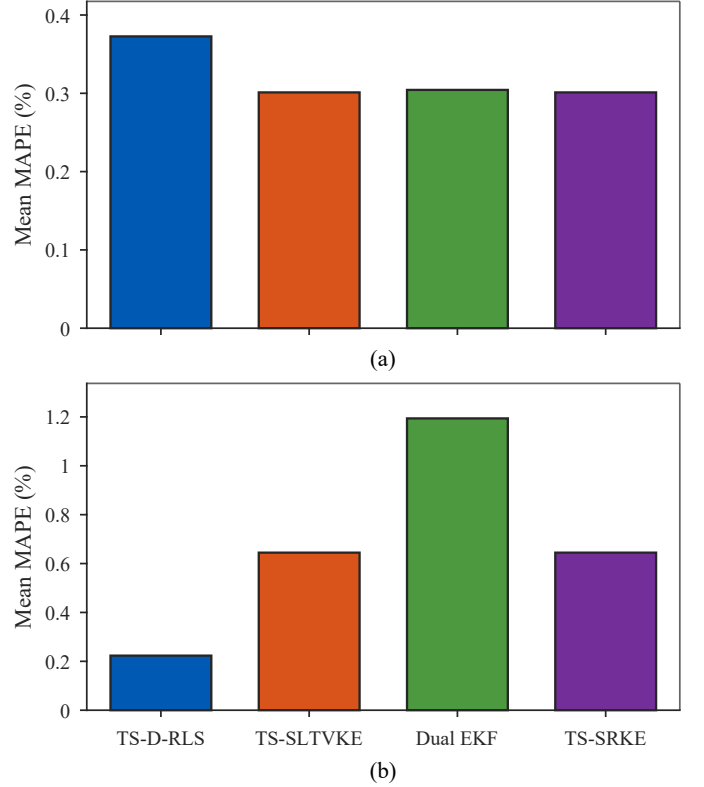


Fig. 6. Static mean MAPE under common O1 observations: (a) capacitance; (b) ESR. The TS-SRKE bars coincide with the TS-SLTVKE bars because the supervisor is silent in steady state.

E. Abrupt Recovery: Resolving the Retained Failure

Abrupt discontinuities are recovery stress tests, not aging events. In the frozen 257-post-step-cycle check (≈ 5.1 ms), the unsupervised TS-SLTVKE still carries 19.2911% error after $C \rightarrow 0.8C$ and 49.9384% after $r_C \rightarrow 2r_C$: once the parameter jumps, every innovation exceeds the NIS gate and the estimate freezes. Over the full horizon (Table VI; settling measured from the health-change instant at the native per-cycle cadence), its capacitance step settles only at 64.72 ms and its ESR and joint steps never recover. TS-SRKE settles the capacitance, ESR, and joint steps in 1.52, 0.48, and 1.20 ms—well within one 20.48-ms external reporting interval—versus 10.72–17.68 ms for TS-D-RLS. At the external reporting cadence, recovery is visible at the first or second reported sample, depending on step-report alignment (Fig. 9). Dual EKF settles the ESR step fastest (0.24 ms) because it carries no gate, but its uncalibrated coverage, worst false-ESR transients, and highest cost are unchanged, so it remains excluded by the predeclared selection criteria of Section VI.

F. Slow Degradation Tracking

Across the 0.1–100 s sweep in Table VII, TS-D-RLS has lower aggregate combined nRMSE and normalized lag at all four durations, and the supervised TS-SRKE tracks its Kalman parent to within rounding: slow drift stays below the supervisor threshold, so the resets that resolve abrupt steps deliberately do not engage on health-realistic ramps

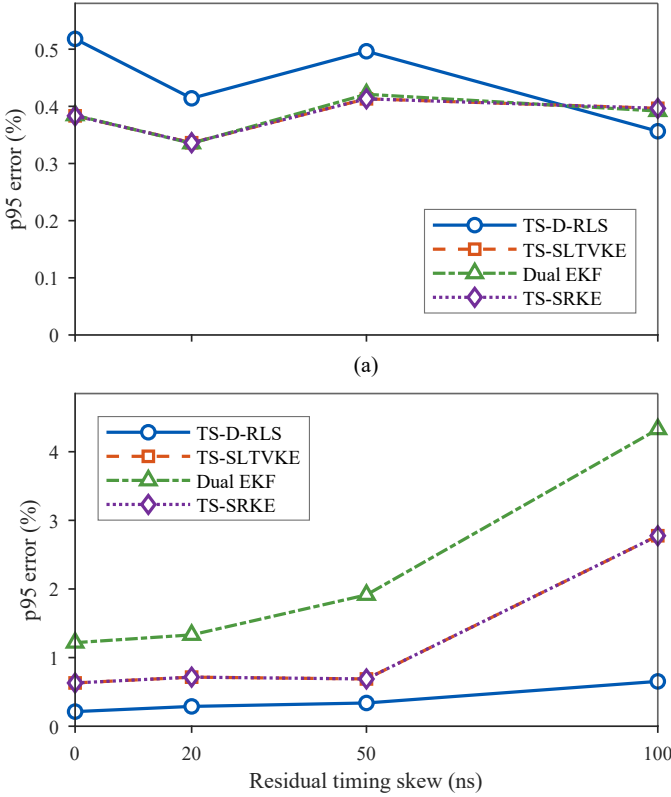


Fig. 7. p95 errors versus residual timing skew under the F28379D-realistic noise profile and shared O1 observations: (a) capacitance; (b) ESR. The TS-SRKE curve overlays its Kalman parent exactly; distinct line and marker styles preserve method identity in grayscale.

TABLE VI
ABRUPT-STEP SETTling ON FROZEN STRESS STREAMS
(FROM THE HEALTH-CHANGE INSTANT, NATIVE PER-CYCLE CADENCE)

Method	$C \rightarrow 0.8C$ (ms)	$r_C \rightarrow 2r_C$ (ms)	joint (ms)
TS-D-RLS	10.72	17.68	17.28
TS-SLTVKE	64.72	no recovery	no recovery
Dual EKF	61.00	0.24	61.36
TS-SRKE	1.52	0.48	1.20

(isolated firings marginally improve the 10-s aggregate). The lightweight baseline therefore keeps a genuine ramp-tracking advantage, reported as such; Fig. 10 shows representative joint trajectories against the planted truth. Individual trajectories can still favor the Kalman side; one 10-s C-ramp case does so.

G. Operating-Point Transient Immunity

Load, input-voltage, and duty transitions hold health fixed. TS-SLTVKE reduces the mean false-C peak (0.7570% versus 1.3017%), whereas TS-D-RLS reduces the mean false-ESR peak (1.8474% versus 5.1919%) and recovers faster. The supervised TS-SRKE reproduces its parent on every transient row—operating transients trigger no reset, so the supervisor adds no false-health response of its own. Neither kernel family dominates operating-transient immunity; Supplementary Fig. S4 shows the retained load-step traces.

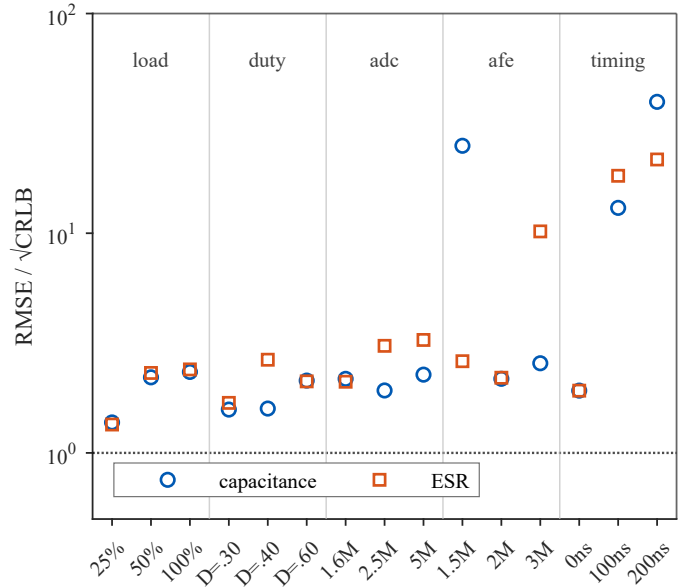


Fig. 8. Empirical RMSE relative to the conditional CRLB over fifteen sensitivity cases (twenty seeds each), grouped by dimension. The bound is conditional on accepted rows and models noise only, so ratios far above one indicate calibration or timing bias rather than estimator inefficiency.

TABLE VII
AGGREGATE SLOW-DEGRADATION TRACKING
(M1 = TS-D-RLS, M2 = TS-SLTVKE, M4 = TS-SRKE)

Ramp (s)	Combined nRMSE			Normalized lag (%)		
	M1	M2	M4	M1	M2	M4
0.1	0.1161	0.2160	0.2160	8.053	22.027	22.027
1	0.0163	0.0454	0.0454	1.360	3.537	3.537
10	0.0160	0.0456	0.0449	1.752	4.492	4.492
100	0.0213	0.0477	0.0478	2.827	4.868	4.868

H. Closed-Loop Operation

All preceding rows keep the duty command in open loop. A supplementary study closes a deliberately simple, low-bandwidth discrete PI voltage loop with 0.5% duty quantization around the Model-A plant and re-commissions the edge gain under feedback ($k_R^{\text{cl}} = 0.9987$); this and the light-load study below evaluate the observation pipeline with the lightweight TS-D-RLS realization. Three blind health cases then apply a $1.45\times$ load step, a $+20\%$ input-voltage step, and a 16-to-12-V reference step at 30 ms. Steady-tail capacitance and ESR errors remain at or below 0.222% and 0.111%, internal convergence takes 3–7 cycles, and the largest post-transition excursions are 0.410% in capacitance and 0.102% in ESR (Fig. 12). During the input-voltage transient, 15.7% of cycles fail the CCM gate and are rejected with the estimate held; accuracy is unaffected. The study uses one noise seed per case and does not evaluate controller design itself.

A second supplement runs the supervised realization itself under the same regulator, reusing the frozen closed-loop edge gain and supervisor constants. Across nominal, $1.45\times$ load-step, and 16-to-12-V reference-step cases the supervisor never fires and false-health peaks stay below 0.50%, so regulator transients do not masquerade as health changes. Under feed-

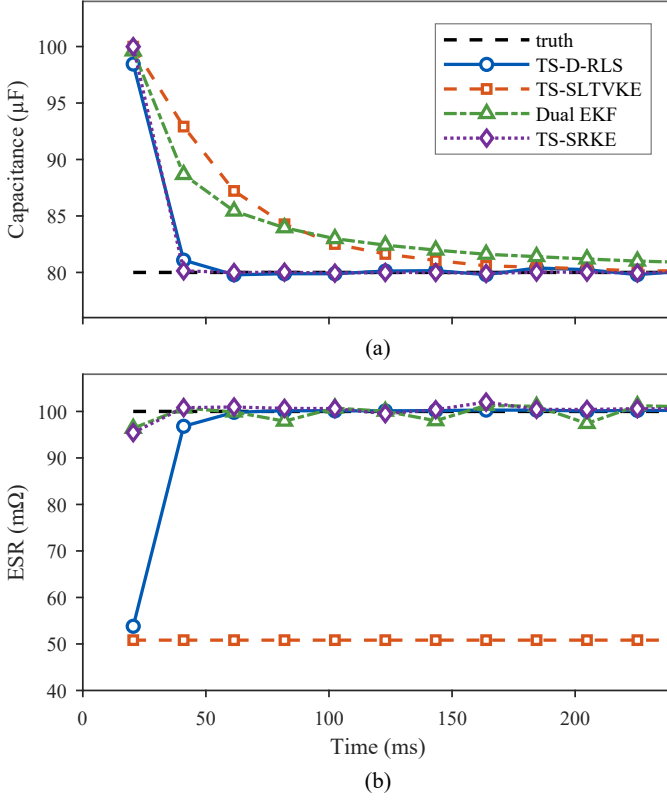


Fig. 9. Report-cadence recovery after the abrupt health steps on identical frozen streams: (a) capacitance estimates after $C \rightarrow 0.8C$; (b) ESR estimates after $r_C \rightarrow 2r_C$. The unsupervised TS-SLTVKE freezes at the stale value; TS-SRKE re-enters the band at the first or second reported sample, depending on step-report alignment.

back, an abrupt $r_C \rightarrow 2r_C$ step fires the supervisor once, ten cycles after the step, and settles in 0.20 ms; a $C \rightarrow 0.8C$ step fires once and settles in 0.28 ms (Fig. 11). The unsupervised parent recovers neither step within the 1500-cycle (30-ms) post-step horizon, freezing at 49.95% ESR error—with 7.69% capacitance error induced through the compensation term—after the ESR step and 19.70% capacitance error after the C step. Supervised tail errors stay at or below 0.26%; each case uses one noise seed.

I. Availability Toward the Light-Load Boundary

Because identifiability is conditional on valid CCM, a fixed-duty sweep quantifies observation availability from 100% down to 6.7% load with the same locked pipeline. Steady-state cycle acceptance remains 100% down to 20% load, falls to 68.8%, 49.8%, 36.1%, and 9.9% at 16.7%, 14.3%, 12.5%, and 10.0% load, and reaches zero at and below 8.3% load, where the estimate is held (Fig. 13). Internal convergence lengthens from 5 to 39 cycles over the fully available range, and endpoint errors over accepted rows stay at or below 1.4% down to 10% load. The switched Model-A equations flag would-be DCM cycles but do not emulate DCM conduction [39]; the sweep therefore quantifies availability, not DCM-mode accuracy.

TABLE VIII
PER-ACCEPTED-OBSERVATION COMPLEXITY

Method	Mult.	Add.	Div.	Est. time (μs)
TS-D-RLS	28	20	1	1.36
TS-SLTVKE	46	34	1	2.12
Dual EKF	62	47	2	3.05
TS-SRKE	50	41	1	2.30

TABLE IX
F28379D DEVICE-REALISTIC SUMMARY

Item	Specified value
ADC architecture	four modules, 16-bit differential
Aperture/rate	320 ns / 1.093 MS/s per module
Edge acquisition	0.5-μs guard; $T_{w,R} = 2.2$ μs; 3 points/side
AFE boundary	differential path; edge-band CMRR ≥ 70 dB
Health reporting	1024 cycles (20.48 ms)
Worst p95 C/ESR	1.6425% / 2.6109%

J. Complexity and Uncertainty

On the common arithmetic basis in Table VIII, TS-D-RLS, TS-SLTVKE, Dual EKF, and TS-SRKE require 28, 46, 62, and 50 multiplications per accepted observation; the supervisor costs four multiplications and about seven additions/comparisons on top of its kernel. The joint interval coverages, against the $\approx 90.25\%$ independence benchmark of Section VI, are 13.00%, 88.93%, 81.90%, and 88.93%, respectively. TS-SRKE therefore carries its parent's near-nominal simultaneous coverage for four extra multiplications per row, and TS-D-RLS remains the lowest-cost tracking realization; Fig. 14 summarizes the accuracy–complexity tradeoff. The listed times are arithmetic estimates, not measured target worst-case execution time (WCET); acquisition and feature-processing costs are separate.

VIII. F28379D DEVICE-REALISTIC FEASIBILITY

The device-realistic study tests whether the internal ADC architecture can acquire the proposed observations under register-realizable timing and an explicit high-common-mode analog front end. Four synchronized 16-bit differential ADC modules use a 320-ns aperture and sustain 1.093 MS/s per module. A one-edge schedule places three samples on each side of the reconstructed edge within $T_{w,R} = 2.2$ μs after a 0.5-μs guard. Direct ADC connection is infeasible over the simulated common-mode range, so feasibility presumes the specified differential AFE, an edge-band common-mode rejection ratio (CMRR) bound, an external reference and clock, and a bounded calibration residual.

Seven representative conditions with 200 Monte Carlo seeds each yield worst-case p95 capacitance and ESR errors of 1.6425% and 2.6109%; every condition meets its predeclared threshold. Table IX summarizes the evidence for the feasibility claim. Register timing, aperture margin, direct-memory-access traffic, utilization, AFE component assumptions, common-mode extrema, and coverage ranges are retained in Supplementary Note S1.

The 1.36-μs TS-D-RLS value is an estimator-arithmetic estimate, not platform WCET. Firmware compilation, linker

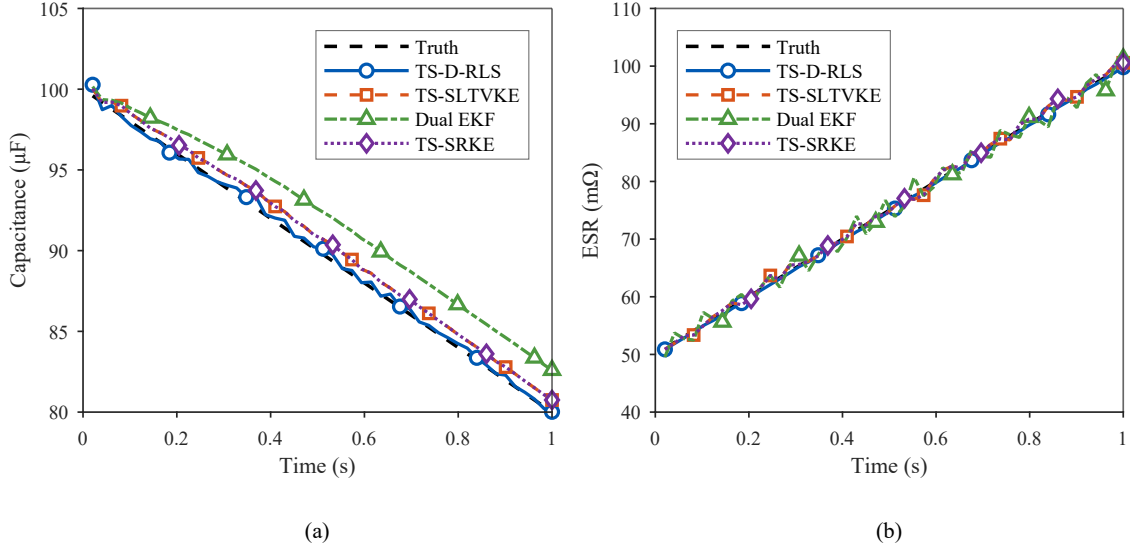


Fig. 10. Representative joint health ramps: (a) capacitance-down and (b) ESR-up trajectories with the planted truth. Long-duration trajectories use trace-derived O1 sufficient statistics rather than full Simscape transients or measured aging data.

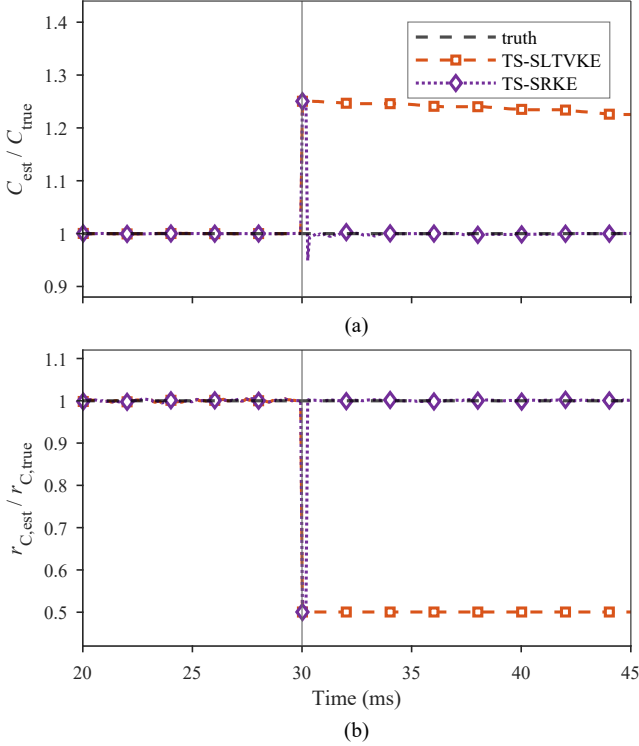


Fig. 11. Supervised recovery under closed-loop regulation on identical noise streams, estimates normalized by the time-varying truth: (a) $C \rightarrow 0.8C$ step; (b) $r_C \rightarrow 2r_C$ step, both at 30 ms (gray line). The unsupervised parent freezes at the pre-step value; TS-SRKE fires once and re-enters the unit line within 0.28 ms.

placement, measured execution time, and bench AFE verification remain open; the result supports a constrained simulation architecture rather than hardware validation.

IX. DISCUSSION

A. Why Observation Design Matters

The O0–O1 factorial separates observation design from estimator complexity: with the RLS kernel unchanged, O1 reduces both MAPE values and the convergence time by the factors reported in Section VII. This attribution agrees with the circuit physics: the switching edge exposes the ESR drop, while the in-state charge balance exposes inverse capacitance. More elaborate recursion cannot reliably compensate for observations that mix these effects or compare mismatched timestamps.

The scoped literature matrix contains no disclosed method combining the Čuk transfer-capacitor target, $+i_{L1} \leftrightarrow -i_{L2}$ commutation, timestamp-reconstructed edge information, and direction-specific charge/edge updates. This is a bounded review finding, not a universal first-in-literature claim.

B. Why TS-SRKE Is Primary and What TS-D-RLS Retains

TS-SRKE satisfies all four predeclared primary criteria: it reproduces its Kalman parent digit for digit on every static, noise/timing, and operating-transient row (zero supervisor firings outside genuine health changes), keeps the parent's 88.93% simultaneous coverage, near the $\approx 90.25\%$ independence benchmark, and settles abrupt health steps within a fraction of one reporting interval where the parent freezes and the RLS baseline needs an order of magnitude longer. The supervisor thus removes the single disqualifying failure of the calibrated realization at four extra multiplications per row. The choice does not imply universal superiority: TS-D-RLS retains the lowest static ESR error, the fastest native convergence, the better aggregate ramp tracking at every duration, and the lowest cost, and it remains the recommended realization when only point tracking under smooth degradation is required. Dual EKF settles the ESR step faster because it

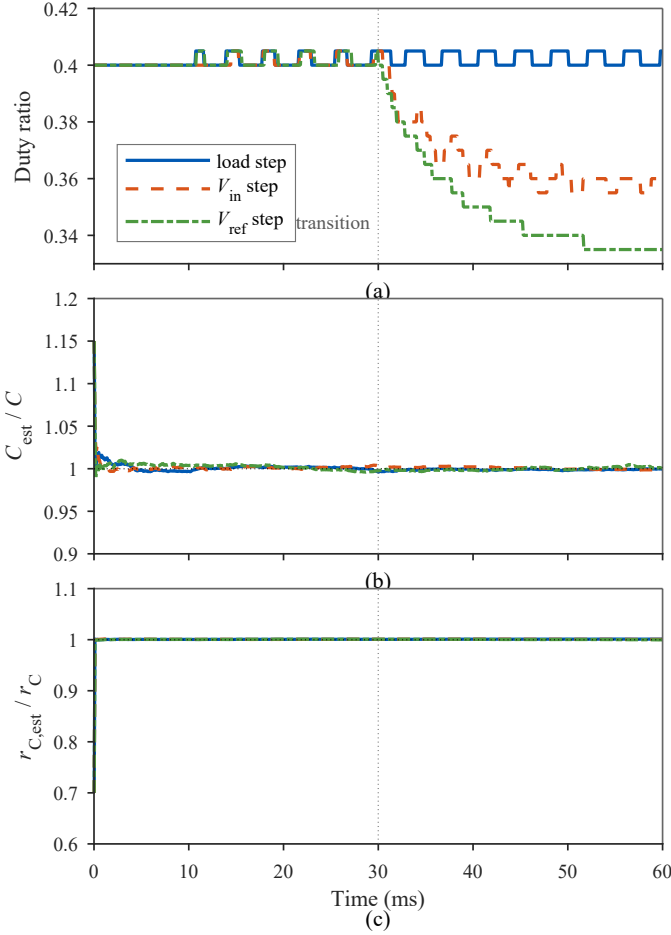


Fig. 12. Closed-loop supplement: (a) regulator duty responses to the load, input-voltage, and reference transitions at 30 ms; (b), (c) normalized capacitance and ESR estimates. Cycles failing the CCM gate during the input-voltage transient are rejected and hold the estimate.

carries no gate, but its uncalibrated coverage and worst false-ESR transients keep it excluded. The two retained realizations are thus complementary—TS-D-RLS as the lightweight practical estimator and TS-SRKE as the uncertainty-aware abrupt-change extension. The minimal supervised closed-loop check of Section VII shows zero false supervisor firings under regulator transients and sub-millisecond recovery of abrupt steps under feedback.

C. Limitations

The evidence has seven principal boundaries.

- 1) Identifiability and updates require valid CCM; DCM and weak-excitation intervals hold the estimate, with availability quantified toward the light-load boundary in Section VII. The method assumes availability of both inductor currents; applications lacking either measurement require an additional sensor or independently validated current reconstruction.
- 2) The reported ESR is an effective condition indicator at the modeled switching, thermal, and frequency conditions, not a temperature-independent aging variable. In deployment, temperature-induced ESR variation can ex-

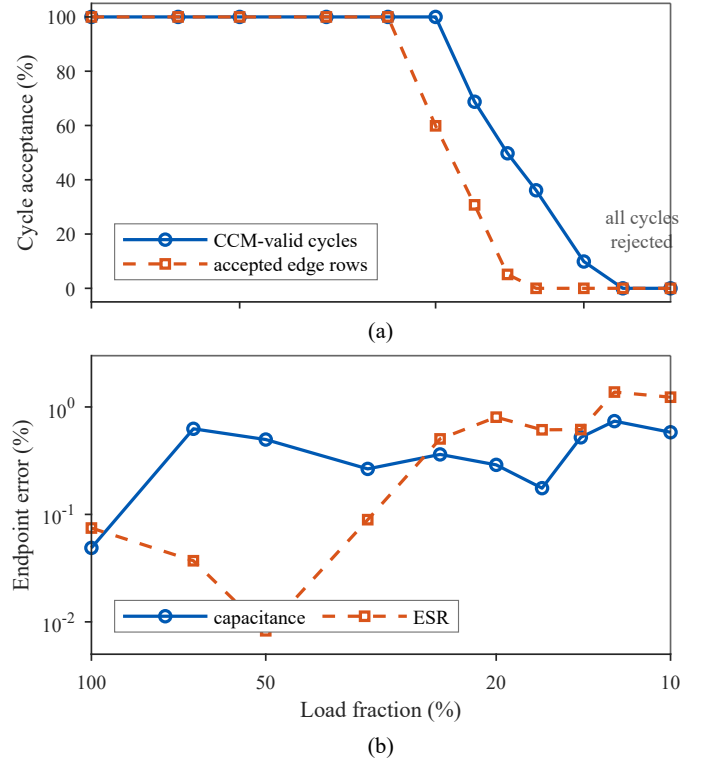


Fig. 13. Light-load availability sweep at fixed duty: (a) steady-state CCM-valid cycle and accepted-edge-row fractions; (b) endpoint errors over accepted cycles. Below the availability knee the gates reject all cycles and the estimate is held.

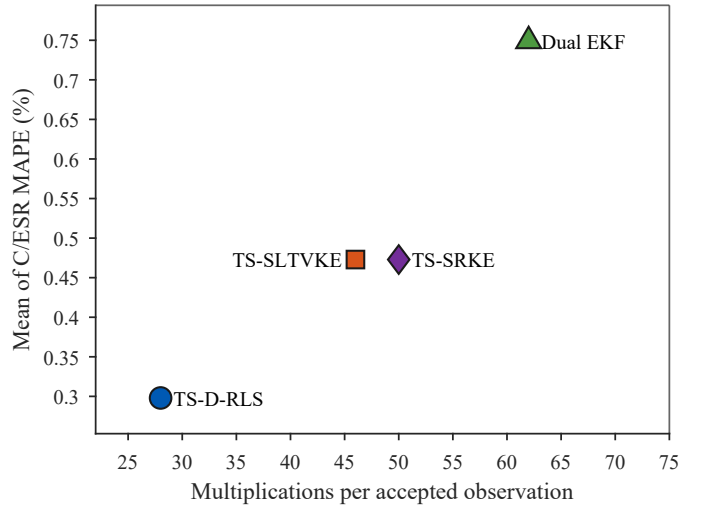


Fig. 14. Accuracy-complexity tradeoff under the common operation-count model. The ordinate is the arithmetic mean of C and ESR MAPE values, used only as a visual Pareto summary and not as the estimator-selection objective.

ceed the identification error reported here; health trending therefore requires thermally matched comparisons or a mapping to a reference temperature $r_C(T_{ref})$, which the simulation scope of this work does not establish.

- 3) The 0.1–100 s trajectories are synthetic or trace-derived bandwidth tests, not run-to-failure data; abrupt discontinuities are recovery stress tests rather than aging events.
- 4) Hardware validation remains open for edge ringing,

isolation/probing, AFE common-mode rejection, and production calibration.

- 5) The F28379D study is device-realistic simulation; target compilation and measured WCET have not been demonstrated.
- 6) The supervisor resolves the abrupt-recovery side of the former tracking–confidence tradeoff, but TS-D-RLS still tracks smooth ramps better and the auxiliary TS-D-RLS intervals remain under-covered; the supervisor’s components are classical, and its constants, while predeclared and shared across all cases, were designed against the evaluated disturbance classes.
- 7) The closed-loop and light-load supplements use the Model-A plant with one noise seed per case and one deliberately simple PI regulator; the supervised closed-loop check shares these limits. They do not evaluate controller-interaction breadth or DCM-mode operation.

D. Experimental Roadmap

The next stage should use a controlled C/ESR network in the energy-transfer branch, the specified F28379D schedule, separate absolute- and edge-voltage paths, and synchronized inductor-current measurements. Tests should sweep input voltage, duty ratio, load, and temperature; characterize ringing and AFE rejection; verify calibration against offline impedance measurements; and measure WCET before firmware release.

X. CONCLUSION

This paper converted the normal bidirectional energy transfer of a Ćuk converter into two topology-synchronous capacitor-health observations—a timestamp-reconstructed switching-edge row for ESR and an ESR-compensated safe-window charge row for capacitance—and proved positive finite-window information bounds that establish local C–ESR identifiability under valid CCM gates. On these observations, the supervised-reset estimator TS-SRKE closed the tracking–confidence gap of the unsupervised kernels: it reproduced its Kalman parent exactly in steady state with near-nominal 88.93% simultaneous interval coverage, and settled abrupt capacitance and ESR steps in 1.52 and 0.48 ms where the parent froze, at four extra multiplications per row.

On the blind matrix, replacing the generic mixed observation with the proposed set reduced capacitance MAPE from 7.0366% to 0.3727% with the RLS kernel unchanged—the observation design, not estimator complexity, carries the largest gain. TS-SRKE then achieved 0.3011%/0.6446% C/ESR MAPE, while the lightweight TS-D-RLS baseline retained the best static ESR error, ramp tracking, and cost, and remains recommended where only smooth-degradation point tracking is required. Supplementary studies preserved pipeline accuracy under closed-loop regulation, quantified availability down to the light-load boundary, and kept noise-driven error within small multiples of the conditional noise-only CRLB at the commissioned configuration; a constrained F28379D simulation supported device-realistic feasibility with worst p95 errors of 1.6425%/2.6109%.

Hardware and accelerated-aging studies must still validate AFE common-mode behavior, ringing, temperature dependence, calibration transfer, and measured execution time before experimental or lifetime claims are warranted.

APPENDIX A

SWITCHED EQUATIONS UNDER THE ADOPTED SIGN CONVENTION

For completeness, the first-order switched equations used to verify the observation signs retain the transfer-capacitor ESR and set the other parasitics to zero. Here v_C is the ideal capacitor voltage and $v_T = v_C + r_C i_C$. With $u = 1$ (main switch on),

$$\dot{i}_1 = V_{in}/L_1, \quad \dot{i}_2 = (v_C - r_C i_2 - v_o)/L_2, \quad (35)$$

$$\dot{v}_C = -i_2/C, \quad \dot{v}_o = i_2/C_o - v_o/(RC_o). \quad (36)$$

With $u = 0$ (main switch off),

$$\dot{i}_1 = (V_{in} - v_C - r_C i_1)/L_1, \quad \dot{i}_2 = -v_o/L_2, \quad (37)$$

$$\dot{v}_C = i_1/C, \quad \dot{v}_o = i_2/C_o - v_o/(RC_o). \quad (38)$$

These equations imply (2) and match the specified Model-A implementation after its optional inductor, switch, diode, and source parasitics are set to zero. The ESR signs differ between states because $i_C = +i_1$ for $u = 0$ and $i_C = -i_2$ for $u = 1$.

APPENDIX B

SCALAR COVARIANCE ENVELOPE

Let the posterior scalar variance at the start of a finite window be P_n . Process accumulation gives $P_n^- \leq P_n + Q_N$. Assimilating total scalar information μ gives

$$P_{n+1} \leq \left(\frac{1}{P_n + Q_N} + \mu \right)^{-1}, \quad (39)$$

which is (25). Setting $P_{n+1} = P_n = P^*$ gives

$$\mu(P^*)^2 + \mu Q_N P^* - Q_N = 0, \quad (40)$$

whose positive root is (26). The map is increasing and bounded above by $1/\mu$. When $Q_N = 0$, inversion gives $P_{n+1}^{-1} \geq P_n^{-1} + \mu$ and hence $P_n \leq (P_0^{-1} + n\mu)^{-1}$. With bounded conditional mismatch, a finite case-dependent error term must be added.

The largest validated fixed points are 0.0112191 in the normalized-capacitance direction and 8.82437×10^{-6} in the ESR direction.

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